Sapienza Università di Roma

Department of Mechanical and Aerospace Engineering

Homework 3 — Attitude Control of a Launch Vehicle in Atmospheric Flight

Dynamics and Control of Launch Vehicles

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1 Introduction

This report addresses the design of an attitude control system for a launch vehicle (LV) during the atmospheric phase of flight, at the instant of maximum dynamic pressure ($t = 72$ s, max-$\bar{q}$). At this condition the LV is the hardest to control: the aerodynamic moment destabilises the airframe while the wind disturbance is largest. The design is carried out entirely with *classical* frequency-domain tools, using the Nichols chart as the primary stability indicator, following the short-period pitch-plane model of Greensite [1].

1.1 Plant model

The pitch-plane short-period dynamics are described by the linear time-invariant model (assignment Eq. 1), with state vector $\mathbf{x} = [\,z,\ \dot{z},\ \theta,\ \dot{\theta},\ \eta,\ \dot{\eta}\,]^\top$ (lateral drift, pitch attitude and first bending mode), control input δ (TVC deflection) and disturbance α_w (wind angle of attack):

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & a_1 & a_1 V + a_4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & A_6/V & A_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\omega_1^2 & -2\zeta_1\omega_1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ a_3 \\ 0 \\ K_1 \\ 0 \\ -\phi_{1,TVC} T_c \end{bmatrix} \delta + \begin{bmatrix} 0 \\ -a_1 V \\ 0 \\ -A_6 \\ 0 \\ 0 \end{bmatrix} \alpha_w. \quad (1)$$

The pitch channel $\ddot{\theta} = A_6\,\theta + K_1\,\delta - A_6\,\alpha_w$ is the core of the problem. The coefficient $A_6 = N_\alpha l_\alpha / I_{yy} > 0$ (*aerodynamic moment*, denoted μ_α) places an open-loop pole at $s = +\sqrt{A_6} = +1.84$ rad/s in the right half-plane: the airframe is **statically unstable** and feedback is mandatory. The coefficient $K_1 = T_c l_c / I_{yy}$ (*control effectiveness*, μ_c) sets the authority of the TVC.

The sensed quantities are produced by the inertial unit (assignment Eq. 2), which couples the bending coordinate η into the gyro and accelerometer channels through $\sigma_{1,ins}$ and $\phi_{1,ins}$:

$$\theta_m = \theta + \sigma_{1,ins}\,\eta, \quad \dot{\theta}_m = \dot{\theta} + \sigma_{1,ins}\,\dot{\eta}, \quad z_m = z - \phi_{1,ins}\,\eta, \quad \dot{z}_m = \dot{z} - \phi_{1,ins}\,\dot{\eta}. \quad (2)$$

This contamination is exactly what makes the lightly damped bending mode ($\omega_{BM} = 18.9$ rad/s, $\zeta_{BM} = 0.005$) visible to the controller and drives the need for a dedicated filter (Task 2).

The actuator is the second-order TVC of Eq. 3,

$$W_{TVC}(s) = \frac{\omega_{TVC}^2}{s^2 + 2\zeta_{TVC}\omega_{TVC} s + \omega_{TVC}^2}, \qquad \omega_{TVC} = 70 \text{ rad/s},\ \zeta_{TVC} = 0.7, \quad (3)$$

in series with a pure transport delay $\tau = 20$ ms, approximated by a third-order Padé rational function. Parameters at $t = 72$ s are read from the reference data set `GreensiteLPV_DATA.mat` (cross-checked against Table 1 of the assignment); the salient values are reported in Table 1.

Table 1: Pitch-plane LV parameters at the max-$\bar{q}$ condition ($t = 72$ s).

Quantity	Symbol	Value
Aerodynamic moment	$A_6 = \mu_\alpha$	3.382 s^{-2}
Control effectiveness	$K_1 = \mu_c$	4.565 s^{-2}
Drift coefficients	a_1, a_3, a_4	$-0.0154,\ 20.61,\ -27.27\ \mathrm{s}^{-2}$
Relative velocity	V	937.7 m/s
Bending mode	ω_{BM}, ζ_{BM}	18.9 rad/s, 0.005
INS coupling	$\phi_{1,ins}$ [–], $\sigma_{1,ins}$ [rad/m]	0.8, 0.178
TVC actuator	$\omega_{TVC}, \zeta_{TVC}, \tau$	70 rad/s, 0.7, 20 ms

1.2 Control architecture

A proportional–derivative attitude law is adopted, augmented by a weak feedback on the lateral-drift channel:

$$\delta_{cmd} = K_{P,\theta}\,(\theta_{ref} - \theta_m) - K_{D,\theta}\,\dot{\theta}_m - K_{P,z}\,z_m - K_{D,z}\,\dot{z}_m, \tag{4}$$

with $K_{P,z}, K_{D,z}$ small and negative, of order 10^{-3}, as recommended in the assignment guidelines. Closing the loop also on $(z_m, \dot{z}_m)$ trades between two classical objectives in wind: containing the lateral drift (*drift minimum*) and containing the angle of attack $\alpha = \theta + \dot{z}/V + \alpha_w$, hence the structural load $\bar{q}\,\alpha$ (*load minimum*) — the binding concern precisely at the max-$\bar{q}$ condition studied here. The weak negative gains keep the drift bounded without fighting the natural load-relieving weathercock of the airframe; both z and $\bar{q}\,\alpha$ are therefore monitored in the gust simulations. The actual deflection follows from the actuator and the bending filter $H_x(s)$ (Task 2): $\delta = W_{TVC}(s)\,e^{-\tau s}\,H_x(s)\,\delta_{cmd}$.

1.3 A note on conditional stability

Because the open-loop airframe carries a right-half-plane pole, the loop is *conditionally stable*: closed-loop stability is lost if the loop gain is *reduced* below a threshold, not only if it is increased. Consequently the `margin` routine returns the low-frequency *aerodynamic* gain margin, and the assignment targets $|GM| \approx 6$ dB and $|PM| \approx 30°$ are interpreted in magnitude, as read off the Nichols chart. This is standard practice for the attitude control of aerodynamically unstable launchers [2, 3].

2 Task 1 — Rigid Launch Vehicle

2.1 Rigid model and loop

Neglecting the bending mode, the plant reduces to the four states $[\, z, \dot{z}, \theta, \dot{\theta}\,]$ and the actuator is taken as ideal: $W_{TVC}(s) = 1$ and no transport delay. The open-loop transfer broken at the TVC deflection, in the $1 + L$ convention, is obtained by closing the static PD law of Eq. (4) around the plant:

$$L(s) = K_{P,\theta}\, G_{\delta\to\theta}(s) + K_{D,\theta}\, G_{\delta\to\dot{\theta}}(s) + K_{P,z}\, G_{\delta\to z}(s) + K_{D,z}\, G_{\delta\to\dot{z}}(s), \tag{5}$$

where $G_{\delta\to\bullet}$ are the plant channels from δ to each measurement. The dominant term is the pitch loop $L_\theta(s) = (K_{P,\theta} + K_{D,\theta} s)\, K_1/(s^2 - A_6)$, whose closed-loop characteristic polynomial $s^2 + K_1 K_{D,\theta}\, s + (K_1 K_{P,\theta} - A_6)$ is stable only for $K_{P,\theta} > A_6/K_1 = 0.74$ and $K_{D,\theta} > 0$: the proportional gain must overcome the aerodynamic instability.

2.2 Tuning

The pitch gains were tuned (by a small `fminsearch` wrapper around the margin computation) to meet the assignment targets, while the lateral gains were fixed at $K_{P,z} = K_{D,z} = -10^{-3}$. The resulting design is

$$K_{P,\theta} = 1.98, \quad K_{D,\theta} = 1.40, \quad K_{P,z} = K_{D,z} = -10^{-3}, \tag{6}$$

with margins $|GM| = 6.0$ dB at 0.34 rad/s and $|PM| = 30°$ at 0.21 rad/s. In the pole-placement language of the course notes the retained gains place the dominant pitch pair at $\omega_c = \sqrt{K_1 K_{P,\theta} - A_6} = 2.4$ rad/s — inside the typical 1–4 rad/s control band — with an equivalent damping $\zeta = K_1 K_{D,\theta}/(2\omega_c) = 1.34$, deliberately above the 0.7–0.8 pole-placement range because here the damping is driven by the phase-margin target rather than by a step-response specification. The Nichols chart of Figure 1 shows the curve threading between the two critical points $(-180°, 0\,\mathrm{dB})$ and $(-540°, 0\,\mathrm{dB})$ — the geometric signature of conditional stability — and clearing the 6 dB closed-loop contour; the markers locate the phase and gain crossovers where the margins are read.

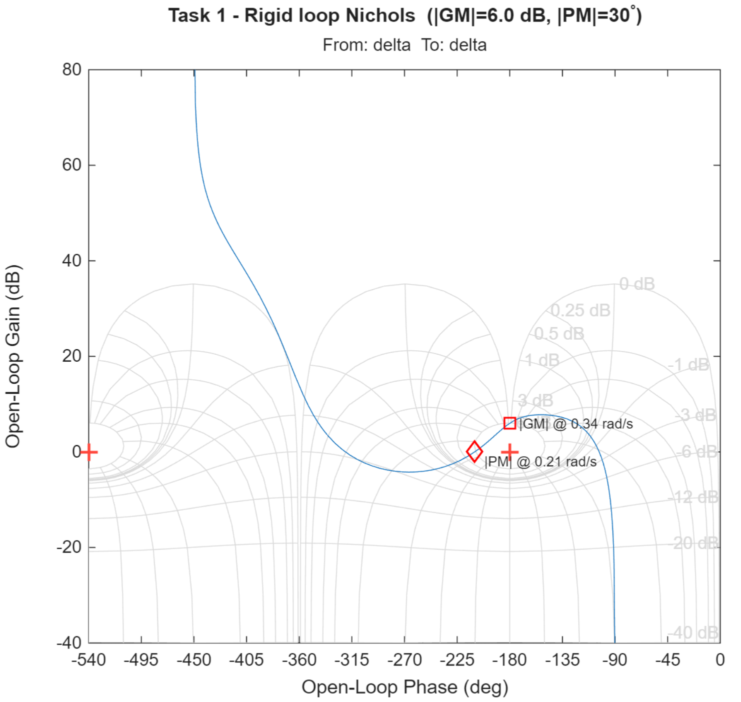

Figure 1: Task 1 — Nichols chart of the rigid open-loop transfer. The loop is conditionally stable; the design meets $|GM| \approx 6$ dB and $|PM| \approx 30°$ (markers at the crossover frequencies).

2.3 Wind-gust response

The closed loop is excited by a deterministic "1-cosine" wind gust whose amplitude ($V_g = 6.4$ m/s) is taken from the reference dry-wind dispersion (`drywind.mat`, severe regime) at the current altitude, converted to a wind angle of attack $\alpha_w = v_w/V$ with a peak of 0.39°. The four key time histories requested by the assignment are shown in Figure 2: the pitch attitude peaks at 0.139°, the TVC deflection at 0.38°, and the lateral drift builds up smoothly to 2.5 m, as expected from the weak drift feedback. The response is well damped and the actuator effort is modest.

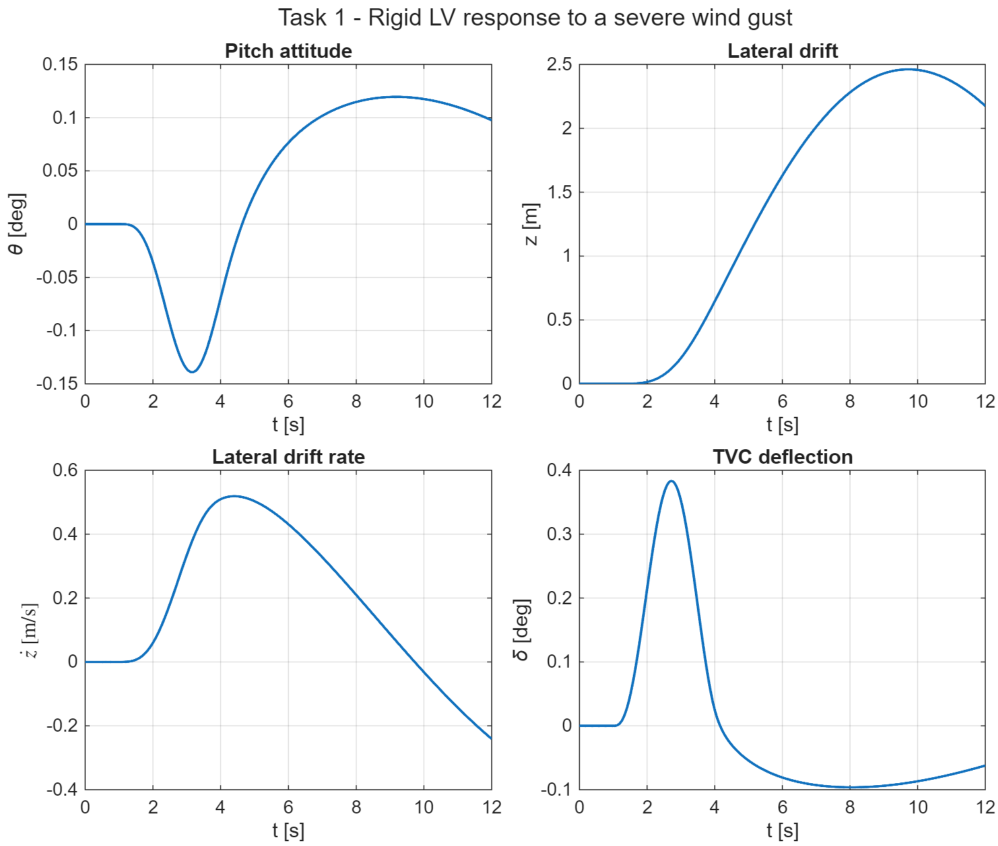

Figure 2: Task 1 — response to a severe wind gust: pitch attitude θ, lateral drift z, drift rate $\dot{z}$ and TVC deflection δ.

Since max-$\bar{q}$ is the load-critical point of the trajectory, the total angle of attack $\alpha = \theta + \dot{z}/V + \alpha_w$ and the structural-load indicator $\bar{q}\,\alpha$ are also monitored (Figure 3). The attitude loop pitches the vehicle slightly into the wind ($\theta \to -0.14°$ at the gust peak), so the peak total incidence, 0.31°, stays *below* the wind contribution alone — a mild load-relieving action. With $\bar{q} = 81$ kPa the peak load indicator is $\bar{q}\,\alpha = 25$ kPa deg, far from typical structural allowables, so at this gust level the drift-minimum–oriented gain choice does not conflict with the load-minimum requirement.

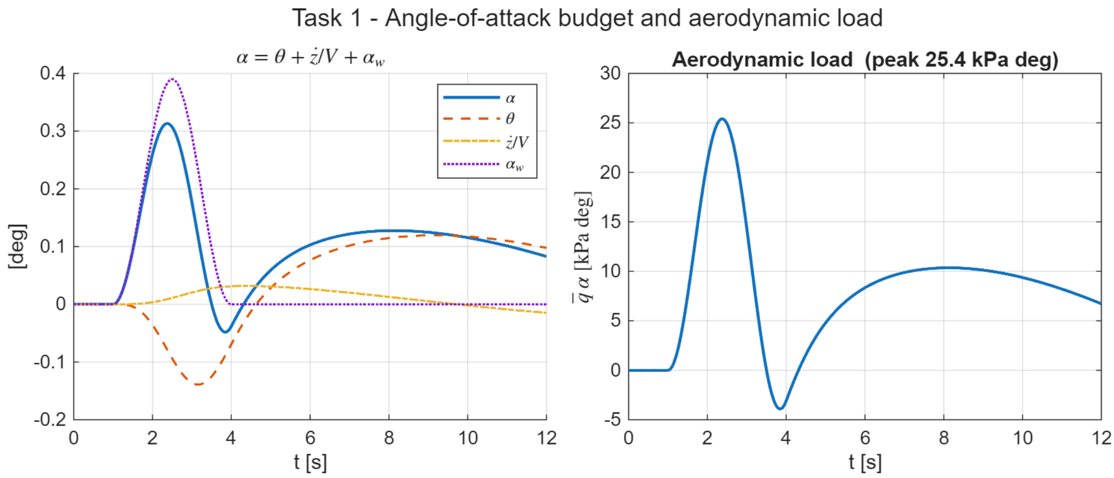

Figure 3: Task 1 — angle-of-attack budget $\alpha = \theta + \dot{z}/V + \alpha_w$ (left) and aerodynamic load indicator $\bar{q}\,\alpha$ (right) during the severe gust.

3 Task 2 — Full Launch Vehicle Model

Task 2 reintroduces the full dynamics of Eq. (1): the lightly damped first bending mode, the second-order TVC actuator of Eq. (3) with its 20 ms delay, and the INS measurement model of Eq. (2). The design proceeds incrementally, as suggested in the assignment guidelines, with the PD gains of Task 1 kept fixed throughout (the guideline note anticipates that retuning is usually unnecessary, and this is confirmed below).

3.1 Step B — uncompensated full model

Adding the TVC, delay and bending mode to the Task 1 controller leaves the margins at low frequency essentially intact, but the bending resonance appears in the loop at a magnitude of $|L(\omega_{BM})| = +39$ dB. With a damping of only $\zeta_{BM} = 0.005$ this peak encircles the critical point and the closed loop becomes **unstable** (a closed-loop pole at $\mathrm{Re}(s) = +5.6$). The red curve in Figure 4 shows the bending loop crossing into the unstable region.

3.2 Step C — bending filter trade study

The guideline filter of Eq. 4 admits two readings: as printed (negative numerator damping, a non-minimum-phase *lead-lag* that phase-stabilises the resonance by rotating it on the Nichols chart) or in its minimum-phase variant (a band-reject *notch* that gain-stabilises the resonance by driving it below 0 dB). Rather than committing a priori, four candidates are compared with the same fixed PD gains; Table 2 collects the resulting margins (rigid-body and minimum gain margin, phase margin, delay margin on top of the 20 ms already modelled, and the residual loop gain at the bending frequency).

- **C-LL — Eq. 4 lead-lag alone.** The filter is swept over the ranges suggested in the assignment ($\zeta_{x,N} \in [0.1, 0.3]$, $\zeta_{x,D} \in [0.4, 0.6]$, $\omega_x = \omega_{BM} \pm 4$ rad/s, 75 combinations): *none* stabilises the loop. The least-unstable combination ($\omega_x = 18.9$, $\zeta_{x,N} = 0.10$, $\zeta_{x,D} = 0.60$) still leaves a closed-loop pole at $\mathrm{Re}(s) = +0.16$ — the green curve in Figure 4 shows its bending lobe still pinned to the critical region. Phase rotation alone cannot rescue a +39 dB resonance here.
- **C-N — deep minimum-phase notch.** Gain stabilisation needs more than 39 dB of attenuation at ω_{BM}, which forces the numerator damping far below the suggested range: $\zeta_{x,N} = 0.002$, $\zeta_{x,D} = 0.7$ give a -51 dB null and drive $|L(\omega_{BM})|$ to -12 dB. The null is deliberately narrow because the bending frequency sits only a decade above the rigid crossover ($\omega_{BM}/\omega_c \approx 8$): a wide band-reject section would leak phase lag into the conditionally stable low-frequency region.

- **C-T — notch triplet.** The course-notes recipe against bending frequency uncertainty places three notches at $\{0.9, 1.0, 1.1\}\,\omega_{BM}$. Sized to the same depth, however, the three wide ($\zeta_{x,D} = 0.7$) sections pile up about 30° of phase lag at the rigid crossover and the conditionally stable loop goes *unstable at nominal* ($|PM|$ collapses to 11.6°). This is consistent with the course philosophy: the triplet belongs to the higher, well-separated modes, while the *first* mode — too close to the control band — must be treated either with an exact notch or by phase stabilisation.

- **C-NLL — notch + lead-lag.** Reading the guideline literally ("the Notch filter and possibly other filters"), the deep notch is cascaded with the Eq. 4 lead-lag, the partner swept over the same ranges: 31/75 combinations now stabilise the loop. The best one ($\omega_x = 22.9$, $\zeta_{x,N} = 0.10$, $\zeta_{x,D} = 0.40$, selected for maximum delay margin) is stable but with visibly thinner nominal margins ($|PM| = 9.2°$, $\min|GM| = 0.7$ dB). Its merit appears in the robustness analysis below.

Table 2: Task 2 — bending-filter trade with fixed Task-1 PD gains.

Candidate	rigid $\|GM\|$ [dB]	min $\|GM\|$ [dB]	$\|PM\|$ [°]	DM [ms]	$\|L(\omega_{BM})\|$ [dB]	stable
B (no filter)	6.34	6.34	30.8	—	+38.8	no
C-LL (lead-lag)	6.58	0.61	9.0	—	+23.2	no
C-N (deep notch)	6.58	1.81	26.5	58.8	−12.1	**yes**
C-T (triplet)	7.07	1.48	11.6	—	−46.0	no
C-NLL (notch+LL)	6.72	0.73	9.2	21.9	−18.3	yes

The **deep notch is retained**: it gain-stabilises the resonance while preserving the rigid-body margins ($|GM| = 6.6$ dB, $|PM| = 26.5°$ — a few degrees below the 30° of the ideal-actuator loop of Task 1, the difference being the phase lag of the second-order TVC and of the 20 ms transport delay), leaves a 58.8 ms delay margin, and requires no retuning of the PD gains, consistent with the guideline note. The bending mode itself is left at its lightly damped but stable open-loop location. One caveat is recorded: the minimum gain margin of the stabilised loop, governed by the bending shoulder, is 1.8 dB — adequate for this exercise but well below the ≥ 6 dB customarily required of gain-stabilised modes in industrial practice, where the deeper margins would be bought with a stiffer notch or a retuned loop gain.

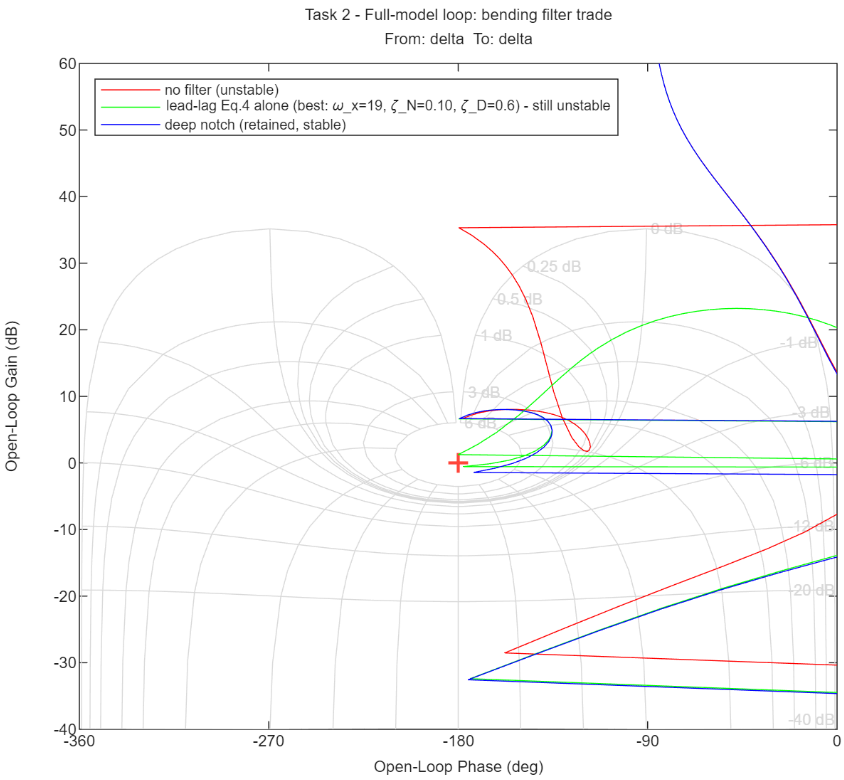

Figure 4: Task 2 — bending-filter trade on the Nichols chart (phase wrapped): without compensation the +39 dB resonance destabilises the loop (red); the best Eq.-4 lead-lag alone rotates the bending lobe but cannot clear the critical region (green); the deep notch gain-stabilises it and restores the rigid margins (blue).

3.3 Sensitivity to the bending-frequency knowledge

The deep notch buys its performance with a -51 dB null only 0.08 rad/s wide, i.e. with the assumption that ω_{BM} is known exactly — true in this exercise, where the LPV dataset provides it, but never true in practice, which is precisely why the course notes prescribe the $\pm 10\%$ triplet. Table 3 makes the trade explicit: each filter is held fixed while the true plant bending frequency is perturbed. The notch tolerates barely -5%; the notch+lead-lag combination, by phase-stabilising whatever leaks past the (detuned) notch, recovers nearly the full $\pm 10\%$ band at the price of the thin nominal margins seen above. In a flight design, where ω_{BM} drifts with propellant depletion, the robust variant (or a retuned phase-stabilised loop) would be mandatory; for the present max-$\bar{q}$ snapshot the exact notch is retained.

Table 3: Task 2 — closed-loop stability with the filters fixed and the true bending frequency perturbed by up to ±10%.

Candidate	$0.90\,\omega_{BM}$	$0.95\,\omega_{BM}$	ω_{BM}	$1.05\,\omega_{BM}$	$1.10\,\omega_{BM}$
C-N — deep notch	no	yes	yes	no	no
C-T — notch triplet	no	no	no	no	no
C-NLL — notch+lead-lag	no	yes	yes	yes	yes

3.4 Validation and comparison with the rigid model

The gust response of the stabilised full model is shown in Figure 5; all four assignment time histories remain well damped, with peaks $\theta = 0.132°$, $z = 2.37$ m, $\delta = 0.39°$ and a load indicator $\bar{q}\,\alpha = 24.9$ kPa deg, marginally below the rigid case. Because the notch is localised at the bending frequency and the TVC bandwidth (70 rad/s) is well above the control crossover, the full-model response is virtually indistinguishable from the rigid one (Figure 6): the added dynamics affect stability robustness (the bending and delay margins) far more than nominal performance.

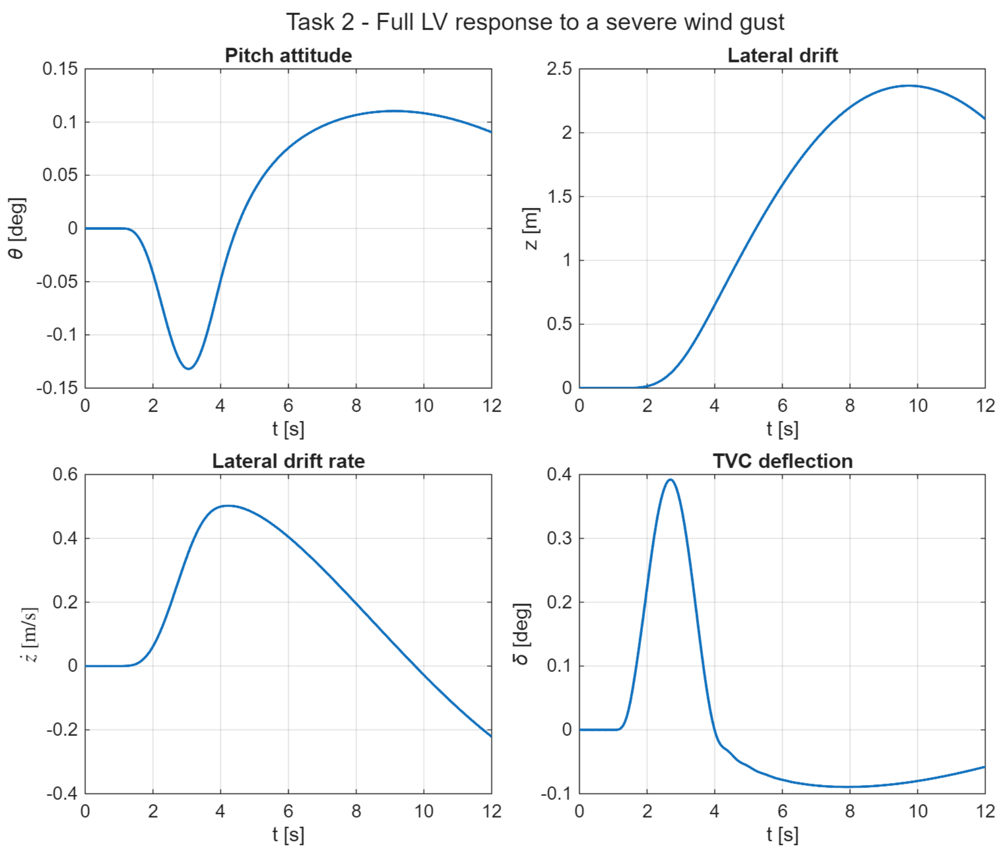

Figure 5: Task 2 — full-model response to the severe wind gust: pitch attitude θ, lateral drift z, drift rate $\dot{z}$ and TVC deflection δ.

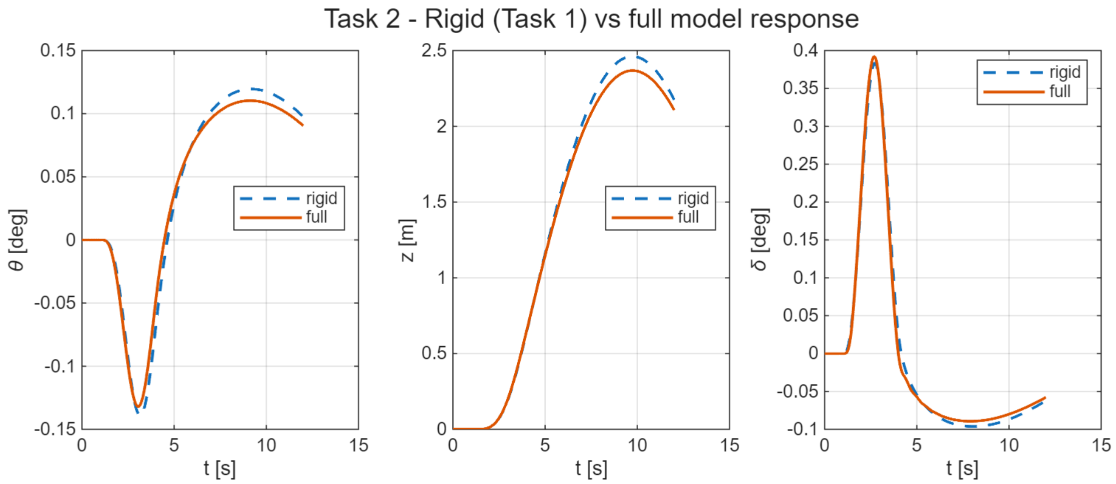

Figure 6: Task 2 — rigid (Task 1) versus full-model gust response: pitch attitude, lateral drift and TVC deflection are nearly coincident.

4 Task 3 — Robustness to Parametric Uncertainty

The aerodynamic moment $\mu_\alpha = A_6$ and the control effectiveness $\mu_c = K_1$ are treated as uncertain. The controller designed in Task 2 (PD gains plus bending notch) is held **fixed** and its robustness is assessed over the *four corner cases* of the uncertainty box, i.e. the vertices V1–V4 obtained by independently varying μ_α and μ_c by $\pm 30\%$. Since each vertex perturbs both parameters at once, the four one-at-a-time variations S1–S4 ($\pm 30\%$ on a single parameter) are also evaluated: they sit on the edge midpoints of the box rather than on its corners, but they cleanly attribute the observed effects to each parameter.

4.1 Frequency- and time-domain analysis

Table 4 collects, for every case, the rigid-body (low-frequency) gain margin, the smallest gain margin over the whole loop (governed by the bending shoulder), the phase margin, the delay margin, and the peak pitch and lateral-drift excursions to the same severe gust. *All vertices remain stable*: the fixed Task-2 controller tolerates the entire $\pm 30\%$ uncertainty box without re-design.

Table 4: Task 3 — margins and gust-response peaks: box vertices (V1–V4, the assignment corner cases) and one-at-a-time sensitivities (S1–S4).

Case	μ_α	μ_c	rigid $\lvert GM\rvert$ [dB]	min $\lvert GM\rvert$ [dB]	$\lvert PM\rvert$ [°]	DM [ms]	peak θ [°]	peak z [m]
Nominal	1.00	1.00	6.58	1.81	26.5	58.8	0.132	2.37
V1	0.70	0.70	6.42	2.81	31.3	134.9	0.136	2.38
V2	0.70	1.30	11.68	0.88	11.3	19.2	0.061	1.67
V3	1.30	0.70	**1.53**	1.53	**9.1**	156.4	**0.353**	**4.20**
V4	1.30	1.30	6.66	0.93	12.3	21.3	0.130	2.36
S1	0.70	1.00	9.44	1.79	25.7	56.1	0.085	1.91
S2	1.30	1.00	4.47	1.83	22.1	61.8	0.187	2.88
S3	1.00	0.70	3.59	2.82	18.6	144.6	0.219	3.16
S4	1.00	1.30	8.79	0.91	11.8	20.2	0.093	2.01

The Nichols overlay over the nominal case and the four vertices (Figure 7) shows the curves clustering at low frequency and fanning out near the bending frequency; the time-domain overlay (Figure 8) shows the corresponding spread of the gust response.

4.2 Discussion

The sensitivities S1–S4 attribute the effects: μ_α moves the unstable airframe pole ($\pm 30\%$ shifts the rigid margin by roughly ∓ 2–3 dB and scales the gust excursions), while μ_c scales the whole loop gain ($+30\%$ improves the rigid margin but pushes the gain-stabilised bending shoulder 2.3 dB closer to 0 dB and halves the delay margin). The vertices then combine these mechanisms:

- **V3** ($\mu_\alpha\uparrow$, $\mu_c\downarrow$) is the *worst case for the rigid loop and for performance*, as the two effects compound: the airframe is more unstable *and* the authority to fight it is reduced. The rigid gain margin collapses to 1.5 dB with $|PM| = 9.1°$, and the gust excursions grow to 0.35° in pitch and 4.2 m of drift — roughly 60% beyond the worst one-at-a-time case (S3), which is why testing the actual vertices, and not only the single-parameter variations, is essential.

- **V2 and V4** ($\mu_c\uparrow$) are the *worst cases for the bending channel*: the loop-gain increase squeezes the minimum gain margin to 0.9 dB and the delay margin to ∼20 ms. The binding constraint in these corners is the flexible side of the loop, not the rigid one.

- **V1** ($\mu_\alpha\downarrow$, $\mu_c\downarrow$) is benign on both counts: the two reductions nearly cancel in the rigid loop and relax the bending shoulder.

In summary, performance degradation is driven by μ_α and mitigated by μ_c, stability of the bending channel is eroded by μ_c, and the worst-case combination V3 leaves single-digit margins. The fixed classical design survives the whole box — but a requirement of, say, ≥ 3 dB and $\geq 10°$ in every corner would already call for either gain scheduling on $\bar{q}$ or a robust-synthesis retune, which is the practical takeaway of this exercise.

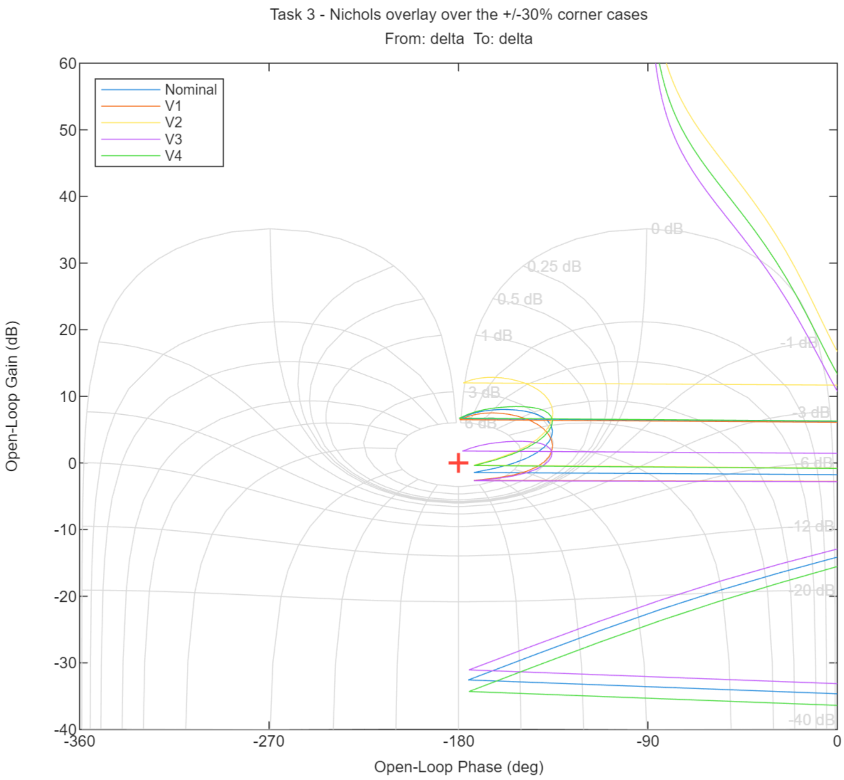

Figure 7: Task 3 — Nichols overlay over the nominal case and the four uncertainty-box vertices.

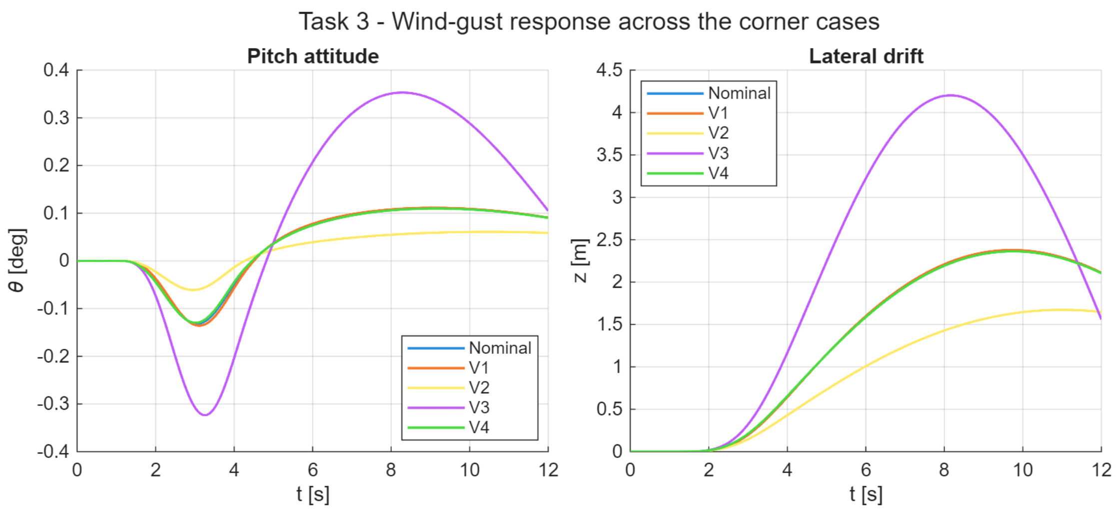

Figure 8: Task 3 — wind-gust response over the nominal case and the four vertices: pitch attitude (left) and lateral drift (right).

5 Conclusions

A classical attitude control system for an aerodynamically unstable launch vehicle at the max-$\bar{q}$ condition was designed and validated entirely in the frequency domain.

- **Task 1.** A proportional–derivative attitude law with weak negative drift feedback stabilises the rigid airframe (open-loop pole at +1.84 rad/s) and meets the design targets $|GM| \approx 6$ dB, $|PM| \approx 30°$. The conditional-stability nature of the loop was identified and accounted for in the interpretation of the margins.

- **Task 2.** Introducing the TVC actuator, the transport delay and the lightly damped bending mode destabilises the loop through a +39 dB resonance. A four-way filter trade showed that the Eq.-4 lead-lag alone never stabilises the loop within the suggested parameter ranges (0/75 combinations), and that a notch triplet sacrifices too much phase at low frequency for a conditionally stable loop; a deep gain-stabilising notch centred on ω_{BM} restores stability while preserving the rigid margins, with negligible change to the nominal gust response. Its reliance on exact bending-frequency knowledge was quantified (±5% detuning is fatal) and a robust notch+lead-lag variant, tolerating nearly ±10% at the price of thinner nominal margins, was identified.

- **Task 3.** The fixed Task 2 controller is robust to ±30% independent variations of μ_α and μ_c: all four vertices of the uncertainty box remain stable. Uncertainty on μ_α erodes the rigid margin and the gust performance, uncertainty on μ_c erodes the bending and delay margins, and the worst vertex (high μ_α with low μ_c) compounds them, leaving only 1.5 dB and 9° — the quantitative argument for gain scheduling in a flight design.

The design illustrates the textbook trade-offs of launch-vehicle attitude control [1, 2]: enough gain to stabilise the unstable airframe, but not so much as to excite the structural modes, with the Nichols chart providing a single, unified view of both constraints. A Simulink block-diagram mirror of the closed loop reproduces the script results and is documented in the accompanying `models/SIMULINK_GUIDE.md`.

References

[1] A. L. Greensite. *Analysis and Design of Space Vehicle Flight Control Systems.* Spartan Books, New York, 1970.

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[3] G. F. Franklin, J. D. Powell, and A. Emami-Naeini. *Feedback Control of Dynamic Systems.* Pearson, Upper Saddle River, NJ, 7th edition, 2015.